

WAKEFIT RELOOP

A Sustainable Product Blueprint for Closed-Loop Mattress Design

From Take – Make – Dispose to a financially viable closed loop

Team- 04

Chirag (015)

Riya Gupta (039)

Diya Singh Rajawat (078)

Mahi Yadav (030)

Saransh Singh (043)

Andeep (008)

The Problem: A Linear Product in a Circular World

Wakefit mattresses, like most of the industry, follow Take → Make → Sell → Use → Dispose. Nothing returns to the value chain.

Figure 1: Current Linear Lifecycle of a Wakefit Mattress (Take - Make - Dispose)



100%

of a mattress's materials — steel, foam and fabric — are lost forever once it is dumped or informally scrapped

3 Pressures

Raw-material price volatility, tightening BIS/EPR regulation, and reputational risk are converging on linear products

Zero Return

No mechanism exists today for steel, foam or fabric to re-enter Wakefit's production once a mattress is sold

Wakefit ReLoop: The Closed-Loop Lifecycle

Take-back → Backhaul → Inspection & Grading → Disassembly → Recovery → back into production.

Take-Back

Bundled with new-mattress delivery to keep collection cost low

Grading

Condition grade (A–C) decides refurbishment, remanufacture or recycling; contaminated units go to cement-kiln co-processing, never landfill

Recovery

Steel, foam and fabric routed to their highest-value recovery pathway

Back to Production

Recovered materials re-enter Wakefit's manufacturing as inputs

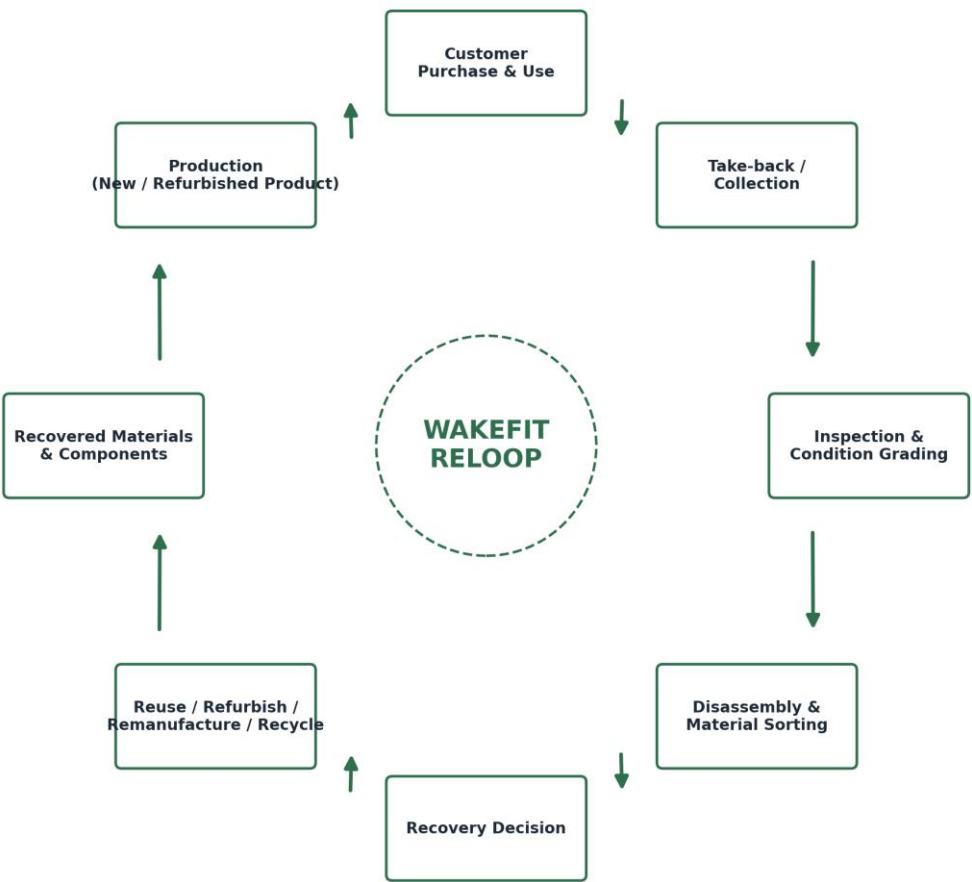


Figure 2: Wakefit ReLoop — Closed-Loop Lifecycle

The Sustainable Product Blueprint

Four strategic quadrants used to design and evaluate Wakefit ReLoop.

1

Material Sourcing & Design

Reduce virgin foam/steel dependence; design for disassembly; standardise spring modules; recycled-content foam.

2

Reverse Logistics

Delivery-linked take-back and exchange programme; inspection, grading, disassembly and sorting.

3

TBL Impact (People · Planet · Profit)

Living wages, ESI & PF for recovery workers; illustrative 100% landfill diversion and 79% material recovery rate; recovered-material and resale value.

4

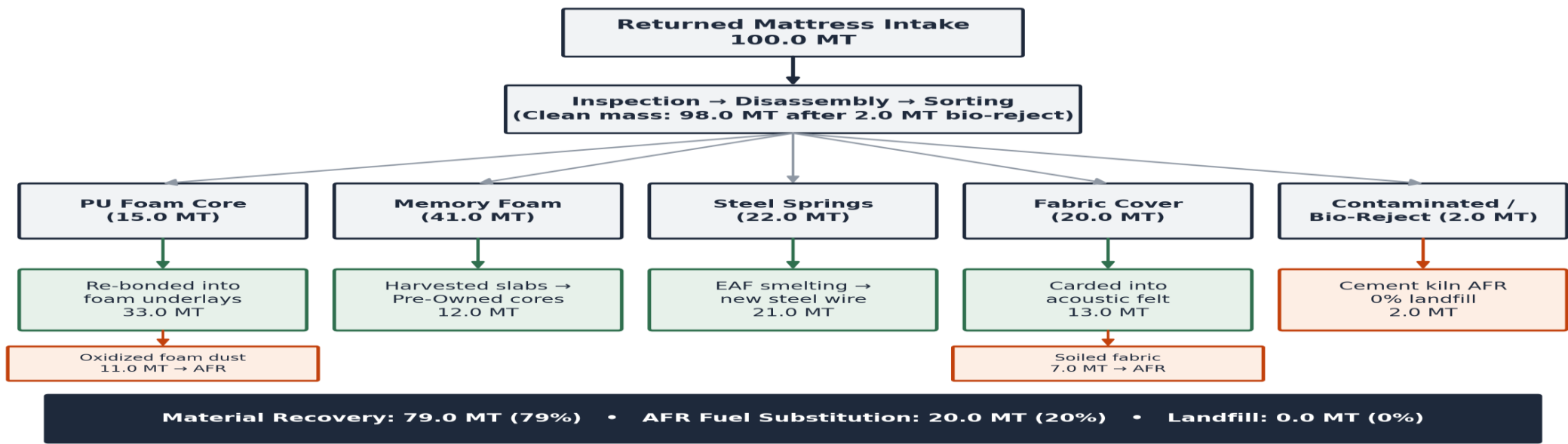
Financial Feasibility

Opportunistic backhaul (~₹450/unit) and trade-in credits protect margin; illustrative unit economics show a net +₹2,224 per mattress, pending pilot validation.

Material Flow & Recovery Pathways

Every returned mattress is inspected, disassembled and sorted into material streams.

Figure 3: Material Flow — Illustrative 100 MT Intake Batch to Recovery Pathway



Illustrative 100 MT intake batch (≈3,330 Queen mattresses): steel springs 22% of mass · PU/memory foam 56% · fabric 20% · rejects/adhesive 2%. Achieves a 79% Material Recovery Rate and 100% Landfill Diversion Rate — subject to empirical validation in the Phase 1 pilot.

People, Planet, Profit — and the Ethical Question

“Who sorts the recycled materials — and are they treated fairly?”

PEOPLE

- Verified fair wages for recovery workers
- Occupational safety standards for cutting/sorting
- Formal contracts, not purely informal labour
- Periodic third-party audits of recyclers

PLANET

- 100% Landfill Diversion Rate (illustrative model)
- Reduced virgin foam & steel demand
- 79% Material Recovery Rate — to be confirmed from pilot batch one
- No zero-waste claim without evidence

PROFIT

- Recovered steel & foam offset input cost — net +₹2,224/unit (illustrative)
- Refurbished-unit resale = new revenue line
- Customer retention via exchange programme
- Must be validated — not assumed — by pilot

The Ask: Pilot Wakefit ReLoop

A data-backed circular model, not a philanthropic add-on — Creating Shared Value in practice.

1

6-month pilot in Bangalore, tied to the Hosur hub

Backhaul pickups restricted to replacement-purchase deliveries, targeting ≤₹500/unit reverse freight

2

Labour standards first

Fair wages, safety and formal contracts fixed before the pilot begins — not after

3

Measure, don't assume

Track Recovery Rate, Diversion Rate and unit economics from batch one

4

Scale on evidence

3-phase national rollout over 36 months, gated on pilot data confirming margin-neutral or better outcomes

Boardroom Test

Pillar 1 — True Circularity: materials genuinely return to production, not just delayed disposal.

Pillar 2 — Economic Viability: steel-first recovery + delivery-linked collection protect margin; full case confirmed by pilot data.

No unprofitable philanthropy. No unverified claims.